

Flow chart :

for each letter in my name,
convert to ascii



divide ascii value by 16 to
obtain hexadecimal value

for birth year, divide birth year by 2 and keep track
of remainder



Keep dividing by 2 until result is 0.

Challenges: I had forgotten how to handle remainders when doing these sorts of conversions though it quickly came back to me.

name = victor

Birth
year = 2003

$$v: 118 \div 16 = 7, v = 76 \\ 6 \div 16 = 0 \quad r = 6$$

$$i: 105 \div 16 = 6, i = 69 \\ \text{remainder} = 9$$

$$c: 99 \div 16 = 6, c = 63 \\ \text{remainder} = 3$$

$$t: 116 \div 16 = 7, t = 74 \\ \text{remainder} = 4$$

$$o: 111 \div 16 = 6, o = 6F \\ \text{remainder} = 15 = F$$

$$r: 114 \div 16 = 7, r = 72 \\ \text{remainder} = 2$$

victor $\Rightarrow$ 76 69 63 74 6F 72

$$2003 \div 2 = 1001 \quad r = 1$$

$$1001 \div 2 = 500 \quad r = 1$$

$$500 \div 2 = 250 \quad r = 0$$

$$250 \div 2 = 125 \quad r = 0$$

$$125 \div 2 = 62 \quad r = 1$$

$$62 \div 2 = 31 \quad r = 0$$

$$31 \div 2 = 15 \quad r = 1$$

$$15 \div 2 = 7 \quad r = 1$$

$$7 \div 2 = 3 \quad r = 1$$

$$3 \div 2 = 1 \quad r = 1$$

$$1 \div 2 = 0 \quad r = 1$$



$$2003 = 1111010011$$